

Manjusha Devi Sunkavalli

Data Analyst

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PROFESSIONAL SUMMARY:

- Highly accomplished Data Analyst with over 5 years of progressive experience in delivering data-driven insights and business intelligence solutions across diverse domains including all domains.
- Demonstrates a deep understanding of data modeling, data architecture, and the full Data Warehouse lifecycle, including ETL processes, data marts, OLAP, and client/server applications.
- Proven expertise in data profiling, cleansing, validation, transformation, and integration to drive actionable insights.
- Expert in crafting and optimizing complex SQL queries for Oracle, Teradata, and Microsoft SQL Server. Proficient in PL/SQL, developing robust stored procedures, functions, triggers, and packages to support high-performance data operations.
- Strong hands-on experience with Informatica PowerCenter and Ab Initio for large-scale ETL processing. Skilled in end-to-end ETL lifecycle from data extraction and transformation to loading and automation, leveraging Unix Shell scripting for job scheduling and automation (including Korn shell).
- Teradata Proficiency: In-depth knowledge of Teradata SQL, utilities (MLOAD, TPUMP, FastLoad, FastExport), and indexing strategies to optimize performance and scalability.
- Performed complex data analysis and profiling across enterprise data sources using SQL, Python (Pandas, NumPy), and Excel. Adept at using Excel Power Pivot, advanced formulas, and VBA Macros for business modeling and KPI tracking.
- Well-versed in designing and validating dashboards and reports using SAP Business Objects (Webi) and Cognos, ensuring alignment with business requirements. Experience testing Business Intelligence reports and validating data across multiple platforms.
- Skilled in querying and mining large datasets to uncover business patterns and trends. Exposure to data mining algorithms, anomaly detection, and exploratory data analysis using Python and SQL.
- Extensive experience in ETL testing, data verification, and ensuring data accuracy and integrity across multiple systems. Well-versed in SDLC and Agile methodologies with strong documentation and test case preparation skills.
- Adept at translating business requirements into functional specifications and data models. Strong communication skills with experience working in onshore/offshore models, collaborating with business stakeholders, project managers, and technical teams.
- Exceptional attention to detail, analytical mindset, and a results-oriented approach.
- Recognized for delivering quality work under tight deadlines and managing competing priorities effectively.
- Strong interpersonal skills and a collaborative spirit, fostering a positive and productive team environment.

TECHNICAL SKILLS:

Databases	Oracle, SQL Server, Teradata, PostgreSQL
ETL Tools	Informatica PowerCenter, Ab Initio
BI Tools	SAP BusinessObjects, Cognos, Tableau (basic), Power BI (basic)
Languages & Scripting	SQL, PL/SQL, Unix Shell Scripting, Perl, Python
OS & Scheduling	Unix/Linux, Control-M, Autosys (basic exposure)
Version Control & Agile Tools	Git, Jira, Confluence
Productivity	Microsoft Excel (Power Pivot, VBA), MS Access, Google Sheets

PROFESSIONAL EXPERIENCE:

Client: Downtown Evening Soup Kitchen, Newhaven, CT

Feb 2025 – Aug 2025

Role: Data Analyst

Responsibilities:

- Interpreted and implemented data mapping specifications to design and execute comprehensive system test plans, ensuring accurate data extraction, transformation, and loading (ETL) from internal data warehouses to external business partners.
- Conducted in-depth analysis of business requirements, system specifications, and data mapping documents while meticulously documenting functional and supplementary requirements using tools like HP Quality Center and JIRA.
- Established and configured test environments based on technical documentation to validate full-system functionality across various scenarios.
- Validated complex ETL mappings and sessions based on business logic to extract data from flat files and relational database systems (RDBMS), ensuring accuracy in data load to target schemas.
- Performed advanced data mining and pattern discovery on healthcare claims datasets using sophisticated SQL queries to uncover trends and support decision-making.
- Executed data mapping activities involving diverse source systems and Teradata for large-scale enterprise data integration.
- Designed and maintained staging environments, loading them with data from heterogeneous sources such as flat files, ASCII, EBCDIC, and RDBMS platforms including Oracle, DB2 UDB, and SQL Server.
- Delivered data outputs in a variety of industry-standard formats, including Excel, CSV, tab-delimited, and pipe-delimited files to meet partner-specific integration requirements.
- Executed targeted marketing campaigns by leveraging customer segmentation criteria and campaign logic defined by stakeholders.
- Adhered to organizational data standardization protocols and governance frameworks to ensure compliance and consistency.
- Performed ad hoc analyses to meet urgent business needs, demonstrating agility in deriving actionable insights from raw datasets.
- Validated XML-based data loads, ensuring successful parsing and loading of structured data into staging databases.
- Authored and executed detailed test cases to verify that source data was accurately transformed and loaded into target systems in the correct format and structure.
- Extensively tested stored procedures, employing complex SQL constructs such as CASE, HAVING, CONNECT BY, and nested subqueries to verify logic and data accuracy.
- Participated in Teradata SQL development, unit testing, and performance tuning to optimize queries and resolve test discrepancies via structured defect triage.
- Validated ETL processes across the full lifecycle—pre-load and post-load—by confirming business rule enforcement, data transformations, and final load accuracy across systems.
- Monitored and verified ETL tool-generated logs and messages, ensuring successful execution and data integrity throughout the data pipeline.
- Created robust UNIX shell scripts for secure file transfers, file parsing, and automated data manipulation tasks, contributing to operational efficiency.
- Advised performance testing teams on appropriate virtual user license allocation, helping to size infrastructure based on anticipated system loads.
- Facilitated onsite-to-offshore transitions, ensuring full knowledge transfer, process alignment, and closure of open defects and issues to maintain QA continuity.
- Performed rigorous database testing, including validation of field lengths, check constraints, and stored procedures, cross-referencing with metadata definitions to ensure structural integrity and business alignment.

Environment: Informatica PowerCenter, DataFlux, Oracle 9i/11g, Teradata, SQL, PL/SQL, TOAD, Quality Center, UNIX Shell Scripting, Flat Files, JIRA, HP ALM, DB2 UDB, SQL Server, Excel, XML, Git, Tableau (basic), Confluence

Client: University Of New Haven, Newhaven, CT

Mar 2024 – Aug 2025

Role: Research Assistant

Responsibilities:

- Conducted cutting-edge research in artificial intelligence, focusing on areas such as natural language processing (NLP), computer vision, and reinforcement learning, contributing to the development of state-of-the-art models and methodologies.

- Developed and fine-tuned transformer-based models (e.g., BERT, GPT, RoBERTa) for tasks including text classification, question answering, summarization, and entity recognition using PyTorch and Hugging Face Transformers.
- Performed extensive literature reviews to stay abreast of current research trends, synthesizing academic papers into actionable insights for experimentation and prototype development.
- Designed and executed experiments to benchmark model performance, leveraging evaluation metrics such as BLEU, ROUGE, F1-score, and perplexity for rigorous model assessment.
- Collaborated with cross-functional teams including data scientists, software engineers, and product managers to align research objectives with business goals and product roadmaps.
- Created custom data annotation pipelines using spaCy and Prodigy to curate high-quality labeled datasets for supervised learning tasks in multiple languages and domains.
- Utilized unsupervised and semi-supervised learning approaches, including clustering algorithms and self-supervised pretraining, to leverage unlabelled data effectively.
- Implemented knowledge distillation and model compression techniques to reduce model size and latency for real-time deployment without significant loss in accuracy.
- Integrated trained models into production environments using Flask, FastAPI, or TorchServe, and monitored performance in real-world applications with tools such as Prometheus and Grafana.
- Applied interpretability techniques such as SHAP, LIME, and attention visualization to demystify model predictions and support decision-making in high-stakes use cases.
- Authored detailed research documentation, technical reports, and white papers to communicate findings and methodologies clearly to both technical and non-technical stakeholders.
- Presented research outcomes at internal seminars, academic conferences, and workshops, contributing to the organization's thought leadership in the AI community.
- Used reinforcement learning techniques (e.g., DQN, PPO) to build agents capable of adaptive learning and decision-making in simulated environments.
- Leveraged large-scale datasets from sources such as Common Crawl, OpenAI APIs, and academic corpora to train and evaluate advanced generative models.
- Explored ethical and responsible AI practices, including bias detection, model fairness evaluation, and privacy-preserving learning (e.g., federated learning, differential privacy).
- Participated in the end-to-end lifecycle of AI research projects, from hypothesis formulation and data sourcing to model deployment and retrospective analysis.

Environment: Python, PyTorch, TensorFlow, Hugging Face Transformers, spaCy, Prodigy, Scikit-learn, NumPy, Pandas, OpenAI APIs, Jupyter Notebook, Flask, FastAPI, TorchServe, Prometheus, Grafana, Git, Docker, Weights & Biases, SHAP, LIME, BERT, GPT, Common Crawl, JSON, YAML, LaTeX, Linux

Client: University Of New Haven, Newhaven, CT

Sep 2024 – Dec 2024

Role: Teaching Assistant

Responsibilities:

- Assisted in the instruction of undergraduate and graduate-level Artificial Intelligence courses, including Machine Learning, Deep Learning, and Natural Language Processing, ensuring students comprehended complex theoretical and applied AI concepts.
- Conducted weekly lab sessions and office hours to provide individualized support, troubleshoot implementation issues in Python (using libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch), and reinforce key learning objectives.
- Designed and graded assignments, quizzes, and exams focused on AI topics such as supervised/unsupervised learning, neural networks, decision trees, and probabilistic models, ensuring alignment with academic standards.
- Developed illustrative Jupyter Notebook tutorials demonstrating real-world applications of machine learning algorithms (e.g., sentiment analysis, image classification, recommender systems) to enhance student understanding.
- Reviewed and provided detailed feedback on student code and project submissions, emphasizing correctness, efficiency, model performance, and adherence to ethical AI practices.

- Collaborated with faculty members to update course content, incorporating current research papers, industry trends, and tools like Hugging Face Transformers, OpenAI APIs, and modern MLOps practices.
- Facilitated group discussions on cutting-edge AI topics such as generative models, reinforcement learning, transfer learning, and model interpretability, fostering an engaging academic environment.
- Managed course infrastructure on platforms such as GitHub, Moodle, and Google Colab, and helped automate assignment collection, evaluation, and plagiarism detection workflows using Python scripts.
- Supported capstone projects and student-led research, mentoring individuals and teams on proposal development, experiment design, data sourcing, model training, evaluation, and final presentation.
- Participated in curriculum development meetings, contributed to the creation of new course modules, and proposed enhancements based on student feedback and observed learning challenges.
- Delivered guest lectures and demonstrations on applied AI topics, including NLP pipelines, model deployment, and the ethical implications of large language models.
- Promoted academic integrity and inclusive learning by ensuring accessibility of materials and offering additional support to students from diverse educational and technical backgrounds.

Environment: Python, Jupyter Notebook, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, OpenAI API, GitHub, Google Colab, Moodle, MS Excel, LaTeX, Google Classroom, Zoom, MS PowerPoint, Google Slides

Client: Cigna, Bangalore, India

Apr 2023 – Jan 2024

Role: Senior Data Analyst

Responsibilities:

- Delivered analytical support on the **Cigna healthcare analytics project** at Accenture, focused on optimizing medical cost containment, utilization management, and patient risk stratification strategies.
- Performed advanced data quality checks and anomaly detection on large-scale claims and EHR datasets using **Python (Pandas, NumPy)** and **SQL Server**, ensuring data accuracy for downstream actuarial and clinical modeling.
- Developed complex **T-SQL scripts** incorporating window functions, CTEs, and stored procedures to extract and transform structured healthcare data for compliance reporting and operational analytics.
- Built **interactive Tableau dashboards** to visualize key healthcare KPIs such as claims cost trends, risk scores, disease prevalence, and patient engagement metrics, improving visibility for Cigna business stakeholders.
- Executed **exploratory data analysis (EDA)** on member demographics, chronic conditions, and utilization data to identify patterns impacting readmission rates, treatment adherence, and cost outliers.
- Applied statistical techniques including **logistic regression, ANOVA, and hypothesis testing** using **R** and **R Markdown** to support predictive modeling and outcomes analysis for population health initiatives.
- Designed and maintained robust **ETL pipelines using SSIS**, integrating data from flat files (CSV, JSON), cloud sources (Azure Blob Storage), and internal data warehouses into SQL Server databases.
- Automated data ingestion, transformation, and validation tasks using **Python scripts**, significantly improving reporting turnaround time and reducing manual processing errors.
- Created ad hoc reports and summary decks in **MS Excel** and **PowerPoint** for Cigna's actuarial, clinical, and business teams, translating technical insights into actionable healthcare recommendations.
- Collaborated cross-functionally with clinical SMEs, business analysts, and compliance leads to align analytics deliverables with evolving healthcare policy and data governance requirements.
- Ensured HIPAA compliance and secure handling of PHI data by adhering to Cigna's data access policies and Accenture's enterprise-grade security frameworks throughout the project lifecycle.

Environment: MS SQL Server, T-SQL, SSIS, Tableau, Python, Pandas, NumPy, R, R Markdown, MS Excel, PowerPoint, VBA, JSON, CSV, SQL Server Management Studio (SSMS), Azure, Windows Server, Jupyter Notebook

Client: Zurich Farmers Insurance Group, Bangalore, India

Feb 2021 – Mar 2023

Role: Data Analyst

Responsibilities:

- Performed end-to-end data analysis on policy, claims, and underwriting data to assess risk exposure, fraud indicators, and loss trends across personal and commercial insurance portfolios at Zurich Farmers Insurance Group.
- Led extensive data quality evaluations and profiling using **Python (Pandas, NumPy)** and **SQL Server**, identifying inconsistencies and ensuring compliance with regulatory and internal data standards.
- Developed and deployed **interactive Tableau dashboards** to track loss ratios, claims frequency/severity, policy churn, and agent performance, empowering underwriters and actuaries with real-time decision support.
- Designed optimized **T-SQL queries** and stored procedures using advanced joins, CTEs, and window functions to efficiently extract and transform large insurance datasets stored in **SQL Server**.
- Executed **exploratory data analysis (EDA)** on claims history, policy tenure, and customer demographics to derive actionable insights that improved pricing models and risk segmentation strategies.
- Utilized **R** for statistical modeling and hypothesis testing (e.g., chi-square tests, t-tests, ANOVA) to evaluate policyholder behavior patterns and identify factors driving claim frequency and premium leakage.
- Authored reproducible **R Markdown reports** combining descriptive analytics and visualizations tailored to stakeholders across underwriting, product, and claims management teams.
- Built and scheduled **ETL pipelines using SSIS** to extract policy and claims data from CSV, Excel, and JSON sources, standardizing ingestion into Zurich's data warehouse.
- Automated pre-processing workflows with **Python**, handling missing data treatment, variable transformation, and dataset merging to support downstream analytics and model readiness.
- Leveraged **Excel** pivot tables, advanced formulas, and VBA macros to prepare and validate ad hoc reports for actuarial and regulatory submissions.
- Collaborated closely with product owners, actuaries, and compliance analysts to define metrics, monitor KPI performance, and enhance the transparency of risk monitoring frameworks.
- Contributed to model governance and documentation efforts, ensuring data lineage, reproducibility, and alignment with Zurich's data privacy and audit standards.

Environment: MS SQL Server, T-SQL, SSIS, Tableau, R, R Markdown, Python, Pandas, NumPy, Microsoft Excel, VBA, JSON, CSV, SSMS, Jupyter Notebook, Windows Server

CERTIFICATIONS:

- Apache Spark Essential Training: Big Data Engineering
- Data Science Foundation: Data Engineering
- MLOps Essentials: Model Deployment and Monitoring
- MLOps Essentials: Model Deployment and Integration
- Apache Kafka Essential Training: Getting Started
- Working with REST and Web Data
- Graph Databases: Neo4j for Complex Data Relationships

EDUCATION:

University of Newhaven, Newhaven, USA

Jan 2024 – Aug 2025

Master of Science, Data Science

Relevant Courses:

Machine Learning & Predictive Analytics(Python), Distributed & Scalable Data Engineering (AWS, ETL Tools),Data Visualization & Storytelling (Power BI), Deep Learning & Computer Vision, Natural Language Processing (NLP), Cloud Computing for Data Applications, Advanced Statistical Methods for Data Science, Math for Data Scientists, Intro to Data Science, Artificial Intelligence, Leadership & Entrepreneurship, R Programming.